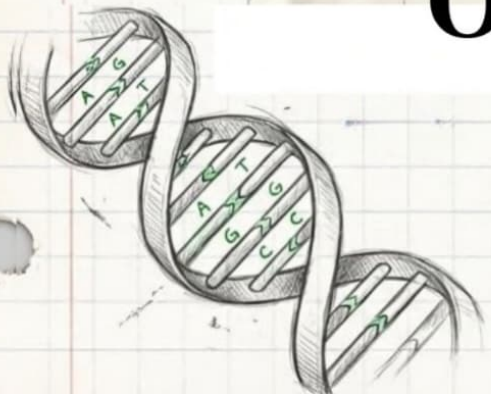
Strictly compiled from expert textbook annotations!

The Molecular Basis Of Inheritance



RNA & The Double Helix Model

RNA Characteristics

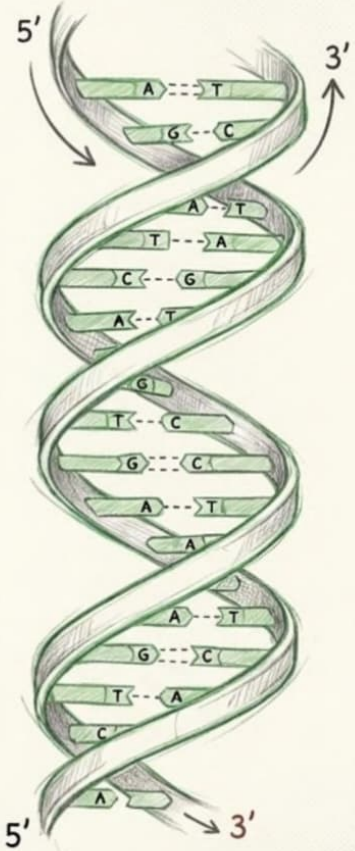
- Has an additional -OH group at 2'-position in ribose.
- Uracil is there at the place of thymine.

The Double Helix Structure:

- James Watson & Francis Crick, based on the X-ray diffraction data produced by Maurice Wilkins & Rosalind Franklin, proposed the Double helix structure of DNA.
- Erwin Chargaff gave a rule that the ratios between A, T, G, and C are constant & equals to one.

The Core Implication:

Since the 2 chains are said to be complementary, therefore if the sequence of bases in one strand is known then the sequence of the other strand can be predicted.



C()

DNA Dimensions & Cellular Packaging

Calculations:

- Distance between 2 consecutive bp = 0.34 nm.
- Total no. of bp in a double helix DNA = 6.6×10^9 bp.
- Total length of DNA = $0.34 \times 10\text{-}9 \times 6.6 \times 10^9 = 2.2$ m.

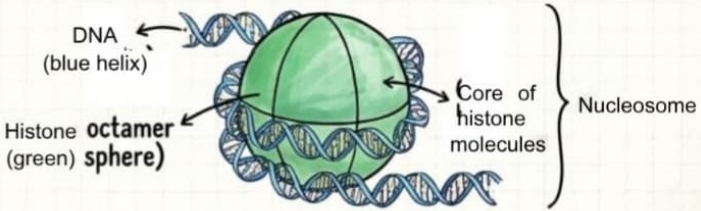
Prokaryotes

- DNA (-vely charged) bound with +vely charged proteins in a region called nucleoid.
- DNA is organised in large loops held by proteins.

Chromatin Types:	
Euchromatin:	loosely packed & transcriptionally active.
Heterochromatin:	tightly packed & transcriptionally inactive.

Eukaryotes(Histones&Chromatin)

- Positively charged basic proteins called histones (rich in Arginine & Lysine). Form a unit of 8 molecules called histone octamers.
- -vely charged DNA is wrapped around histones octamers to form nucleosomes. (A nucleosome contains 200bp of DNA).
- Nucleosomes constitutes the chromatin. Nucleosomes in chromatin seems like "beads on string" structure.
- Get condensed at metaphase stage of cell division to form chromosomes.
- Packaging requires Non-histone chromosomal (NHC) proteins



(A nucleosome contains 200bp of DNA)

The Transforming Principle (Griffith, 1928)

The Setup:

- Streptococcus pneumoniae grown, produces S colonies (smooth) & R colonies (rough).
- S strain bact has mucous (polysaccharide) coat, while R strain does not, thus S strain is virulent.

Step I: S strain given to mice →

it died

Step 2: R strain given to mice →

it lived

Step 3: S strain heat-killed given to mice →

it lived

Step 4: S-strain heat killed + live R strain given to mice →

it died, (living S bacteria recovered from dead mice)

The Conclusion:

- The R strain bacterias had somehow been transformed by S-strain killed bacteria.
- Some 'transforming principle' transferred from heat-killed S-strain bact to R-strain bact which enabled the strain to become virulent.
- Thus, the genetic material in S-strain had somehow transformed the genetic material of R-strain.

Warning: Biochemical nature of genetic material was not defined from his experiments!

Transforming Principle Experiment

The Premise: Oswald Avery, MacLeod and McCarty, the genetic material was thought to be a protein. They worked on the genetic material from the transforming exp.

The Process

Purified biochemicals (proteins, DNA, RNA, etc.) from the heat-killed S cells.

Crucial!

1 | Added Proteases | → Transformation still occurred. Not protein.

| Added RNases | → Transformation still occurred. Not PNA-

Breakthrough!

| Added DNase | → Transformation inhibitted

DNA caused transformation."

Crucial!

The Verdict: They concluded that DNA is the hereditary material.

The Hershey-Chase Exp. (1952)

The Mechanism: Worked with bacteriophages (viruses that affect bacteria). Bacteriophage attaches to bacteria & its genetic mat enters bact cell.

Batch 1 (Radioactive Sulphur)

Viruses grown in S contained radioactive protein because proteins contain sulphur.

Batch 2 (Radioactive Phosphorus)

Viruses grown in P contained radioactive DNA because DNA contains P.

The Blender Step: Radioactive phages were allowed to attach to *E. coli* bacteria, then viral coats were removed from bacteria and they were agitated in a blender.

The Result:

Bacterias infected withviruses having radioactive proteins were not radioactive.

The Result:

Bact infected withviruses thathad radioactive DNA were radioactive, indicating that DNA was the material passed down to bacteria.

Proteins did not enter. DNA is therefore the genetic material!

Properties of Genetic Material: DNA vs. RNA

Context: Stability was very evidentin Griffith's "transforming principle" itself that heat did not destroy some of the properties of genetic material. The two strands being complementary if separated by heating come together, when appropriate conditions are provided.

DNA	RNA
DNA:- More stable / structurally more stable	RNA:-Less stable / labile and easily degradable
DNA: - chemically less reactive	RNA: - Highly reactive (2'-OH group present at every nucleotide is a reactive group)
DNA: - Has Thymine (confers additional stability)	RNA: - Has Uracil
DNA: - Mutate slowly	RNA: - Mutate fastly
Role: Preferred for storage of gen mat. DNA is a better genetic material.	Role: Known to be catalytic. Synthesis of proteins. Preferred for transmission of gen mat.

The RNA World & The Replication Scheme

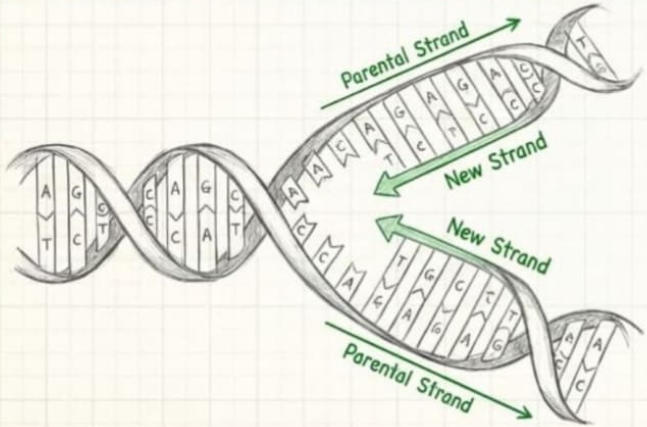
(RNA World:

7 - Used as genetic material & catalyst also.

(-Catalyses many biochem rxns at place of protein enzymes. Thus highly reactive & unstable.

7-DNAevolvedfromRNAwhichwaschem more stable. DNA being double stranded & having

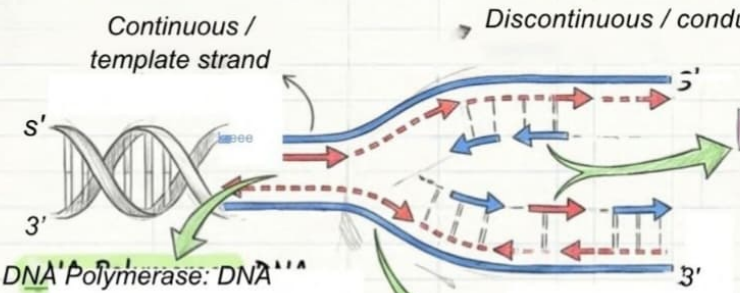
(complementary strands further resists changes by evolving a process of repair.



Replication (Watson &Crick Scheme):

- Watson&Crick proposed this scheme which states that the 2 strands would separate & will act as template strand for synthesis of new strands.
- After completion, there will be a template strand& a newly synthesised strand.

Replication Machinery & Enzymes



DNA ligase: Discont strands are joined by this after their synthesis.

The Process Basics: —

- Origin of replication (ORI): Definite region in E. coli DNA where replication originates.
- In eukaryotes, replication occurs at S-phase of < r cell-cycle. Failure results / into polyploidy. —

DNA Polymerase: DNA dependent DNA polymerase uses a DNA template to catalyse polymerisation of deoxynucleotides. (In E. coli, avg rate is 2000 bp/s & complete replication in 1? mins).

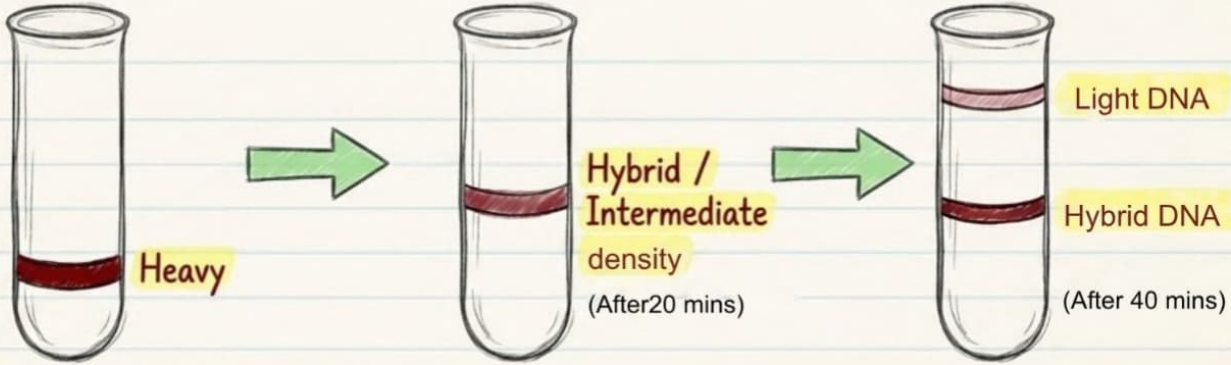
Replication Fork: Since 2 strands of DNA cannot be separated in entire length, replication occurs in a small opening. Polymerases catalyse only in I direction (5'→3').

Deoxyribonucleoside triphosphates: Acts as substrates & provide energy for polymerisation rxn. Its 2 terminal phosphates are high energy. Energetically quite expensive.

Proving Semi-Conservative Replication

The Experiment (Meselson & Stahl):

1. Grew *E. coli* in 25NH4Cl (heavy isotope). ^{15}N incorporated in newly synthesised bact.
2. Grew this newly synthesised *E. coli* in 14NH4Cl (normal isotope).
3. Extracted the DNA so incorporated as double helices struct.



Results: Separated independently in CsCl gradient.

Very similar experiments were performed on Vicia faba!

Transcription:. The Core Principle

Definitions.

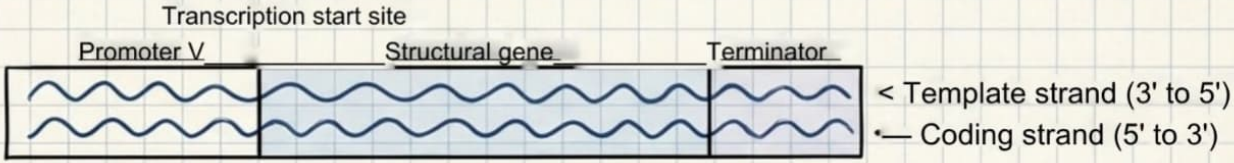
- ORI: It is the ORI where replication starts. It requires a vector to be formed.
- Transcription-. The process of copying genetic info from one Strand of the DNA to RNA.

1 Why both Strands NOT copied i transcription?

I If both strands would act as template, they would code for RNA molecule with diff seq & this would complicate the genetic machinery.

1. 1 RNA molecules if Prod simultaneously would be complementary to each other, thus forming double Stranded RNA, which would prevent it to connect to protein.

The Transcription Unit & The Gene



3 Parts of the Transcriptional Unit:

- (i) Promotor: located towards 5' end (upstream), provides site for binding RNA polymerase & defines Template & coding strands.
- (ii) Structural gene
- (iii) Terminator: located towards 3' end (downstream), provides end of transcription.

Strand Polarity:

- DNA-dependent RNA polymerase catalyses in only one direction (5*—>3').
- Template strand: Polarity 3'->5'.
- Coding strand: Polarity 5'->3'. Has same sequence as RNA (except T instead of U). Strangely, this strand does not code for anything!

Defining a Gene/ Cistron:

- DNA seq coding for tRNA or rRNA define a gene. DNA seq coding for proteins define a cistron. Eukaryotes = monocistronic, Prokaryotes = polycistronic.
- Exons: coding seq present in mature RNA. Introns: do not express, absent in mature RNA.

RNA Types & Eukaryotic Processing

mRNA - Provide template.

tRNA - Brings AA & reads code.

rRNA - Play str & catalytic role.

The Base Process:

- DNA dependent RNA polymerase binds to promotor & initiates transcription. Uses nucleoside triphosphates.

Short stretch of RNA remains bounded. When it reaches terminator, nascent RNA & polymerase fall off. Uses initiation (sigma) & termination (rho) factors.

RNApolymerase iscapableof catalysingthe elongationonly and by associating with initiation(σ)& termination(ρ) factor. In Prokaryotes

In Eukaryotes (3 RNA polymerases in nucleus):

- Pol I: transcribes rRNAs (28S, 18S, 5.8S).
- Pol II: transcribes precursor of mRNA (hnRNA).
- Pol III: Transcribes tRNA, 5srRNA, snRNA.

↓

Processing the Primary Transcript (hnRNA):

Splicing: introns removed, exons joined.

=C=a=p=p=i=n=g=: =m=e=th=y=l guanosine triphosphate added to 5' end.

Tailing: adenylate residues added at 3' end.

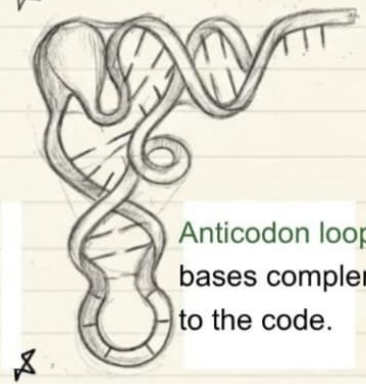
Fully processed mRNA transported out of nucleus for translation.

Primary transcript has both exons & introns, non-functional. Hence, splicing occurs in which introns are removed and exons are joined in defined order.

hnRNA undergoes Capping in which unusual nucleotide (methyl guanosine triphosphate) is added to 5' end & Tailing in which adenylate residues are added at 3' end in a template dependent manner.

Now fully processed hnRNA (i.e., mRNA) transported out of nucleus

tRNA: The Adapter Molecule & Translation Setup*



AA acceptor end: binds to amino acid.

Anticodon loop: has-I bases complementary[1] to the code.

Translation Basics:

- Refers to process of polymerisation of AA to form a polypeptide.
- Formation of polypeptide bonds requires energy.

tRNA Features:

- Specific for each AA.
- There are initiator tRNAs, there are NO tRNAs for stop codons.
- It looks like a compact inverted L.

Charging of tRNA:

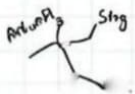
- A A are activated in presence of ATP & linked to their tRNA.
- If 2 charged tRNAs are brought close enough, the formation of polypeptide bond is supported energetically.

Deciphering The Genetic Code

4

The Math & Discoverers:

- GeorgeGamow arguedthat 4 bases coding for 20 amino acids must be a combination of 3 nucleotides generating 64 codons.
- Har Gobind Khorana: chemical method for RNA synthesis.
- Marshall Nirenberg: cell-free system for protein synthesis.
- Severo Ochoa enzyme: polymerising RNA in template independent



Salient Features of the Code:

i) The codon is triplet. 61 code for AA, 3 are stop codons,,

(ii) Some AA coded by >1 codon, hence degenerate,

(iii) Read in mRNA in contiguous fashion (no punctuations).^

(iv) Nearly universal (e.g., UUU = Phenylalanine from bacteria to human).

(v)AUG has dual functions: codes for (Methionine (met)& acts as initiator codon,

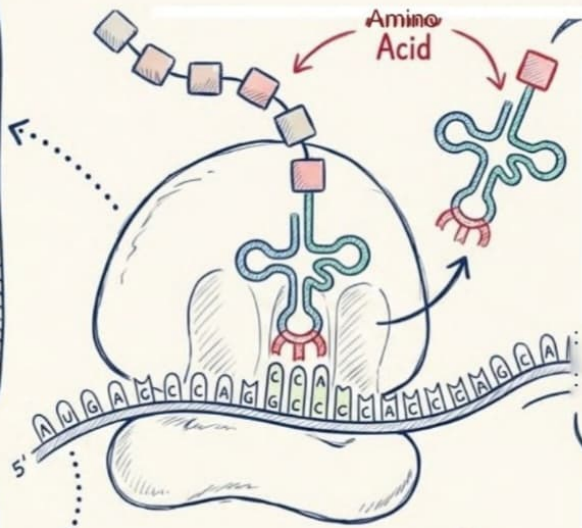
(vi) UAA, UAG, UGA are stop or terminator codons^

The Ribosome's Role:

- Exists as inactive state. When mRNA encounters, translation begins.
- Large subunit has 2 sites for AA to bind close to each other.
- Acts as catalyst (23S rRNA in bacteria is ribozyme) for formation of peptide bond.

Untranslated Regions (UTRs):

- mRNA has UTRs before start codon at 5' end & after stop codon at 3' end.



The Process Steps:

1. Ribosome binds to mRNA at start codon & recognised by initiator tRNA.
2. Continues to form protein chains by complexes (AA linked to tRNA) binding to codon forming base pairs. AA added one by one.)
3. A release factor binds at the end to stop codon, terminating translation & releasing polypeptide from ribosome.

Don't forget:_____

Ribosome = Protein Factory!

X7

J ?22

Regulation of Gene Expression

The genes in cell are expressed to perform a particular function. It is the metabolic, physiological or environmental conditions that regulate the expression of genes.

I Eukaryotic Regulation (4 Levels)

1. Transcriptional level (formation of primary transcript).)
2. Processing level (regulation of splicing).
3. Transport of mRNA from nucleus to the cytoplasm.

1/

Prokaryotic Regulation 7

7 Control of transcriptional initiation is the predominant site for control.

Activity of RNA polymerase at promotor is regulated by accessory proteins. Act positively (activators) & negatively (repressors).

- Accessibility of promotor regions is regulated by interaction of proteins with sequence called operators.

Operators use adjacent to promotors & in many cases bind a repressor protein.

The Human Genome Project (HGP)

13years(1990-2003) | 9billionUSD

The Scale & Scope:

- Would have been 3300 books
- of 1000 pgs req to store info of 3x109 bp. Needed high speed comp, devices.
- J Ass. with Bioinformatics.

Methodologies:

- (i) Expressed Sequence Tags (ESTs): Identify all genes expressed as RNA.
- (ii) Sequence Annotation: Blind approach of seq. all coding & non-coding seq.

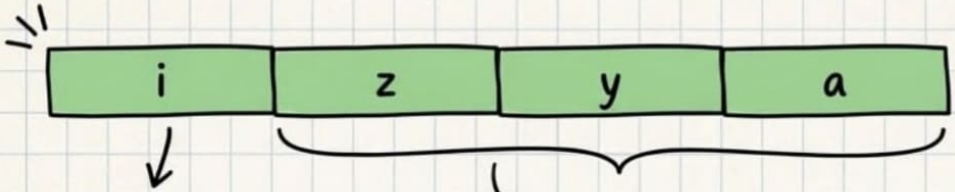
-> DNA isolated —> fragmented -> cloned in host using vectors (BAC, YAC) -> sequenced using Sanger sequencers.

IX Salient Features:

- 1.3164.7 millionbpin human genome.
- 2. Avg gene = 3000 bases; largest is dystrophin (2.4M bases).
- 3. Total genes estimated at 30,000. 99.9%nucleotidebases areexactlythe same inallpeople.
- 4.Functionsunknown for over50%.
- 5. Less than 2% codes for proteins.
- 6.Repeated sequences makeupverylarge portion.
- 7. Chromosome 1 has most genes (2968), Y has fewest (231).
- 8. 1.4 million locations of single nucleotide polymorphism (SNPs).

The Lac Operon & Inducer Mechanism

Found in bacteria. Elucidation by Francois Jacob & Jacque Monod.



1 Regulatory Gene:
codes for repressor.
Synthesised constitutively
(all-the-time).

> 3 Structural Genes:
z = beta-gal (hydrolysis of lactose),
y = permease,
a = transacetylase.

Without Inducer:

Repressor protein binds withoperator region

Prevents RNA polymerase from transcribing.

With Inducer (Lactose):

Repressor is inactivated to interaction with inducer

RNA polymerase accesses promotor. Transcription proceeds!

Sticky Note:
Regulation of lac operon by repressor is referred to as ***negative regulation.***

DNA Fingerprinting & Polymorphism

The Foundation:

- Involves identifying diff in specific reg of DNA seq called repetitive DNA (small stretch repeated many times).
- Separated from bulk genomic DNA via density centrifugations. Small peaks = satellite DNA.
- Don't code for proteins, show high degree of polymorphism. Inheritable from parents to children.

Polymorphism:

--Variation at genetic level arises due to mutations (somatic & germ cells both).

> - If inheritable mutation observed in population at high frequency, it is DNA polymorphism.

The Technique (Alec Jeffreys)

Used VNTR (Variable Number of Tandem Repeats) as probe.

1. Isolation of DNA.
2. Digestion of DNA by restriction endonucleases.
3. Separation of DNA fragments by electrophoresis.
4. Transferring (blotting) of separated DNA fragments to synthetic membranes (nitrocellulose/nylon).
5. Hybridisation using labelled VNTR probe.
6. Detection of hybridised DNA fragments by autoradiography.

Note: Thistechniqueis thebasisofpaternity testing&forensic applicationsdue to highpolymorphism &inheritability! •